MCCT: A Multidimensional Geomorphic Change Detection Strategy for Complex Terrain Areas Using Satellite Remote Sensing Data

Yafei Zhang, Pengfei Li, Liuru Hu, *Member, IEEE*, Jinfei Hu, Hooman Latifi, Bingzhe Tang, Lu Yan, and Pan Zhang

***Abstract*—Accurate monitoring of soil erosion over large areas with complex terrain has faced challenges, which primarily stem from an insufficient capacity of existing satellite remote sensing techniques for monitoring multidimensional geomorphic change in complex terrains. In this study, a framework combining synthetic aperture radar (SAR) and optical satellite imagery was proposed to overcome the above challenges. A multimodal high-resolution satellite image registration method was proposed to mitigate the geometric distortions of high-resolution optical images induced by terrain undulations within small areas in the complex terrain. The registered optical images were employed to derive horizontal geomorphic changes using the optical pixel offset tracking (POT) method, while the surface displacement projected onto the radar line of sight (LOS) was retrieved using multitemporal interferometric SAR (MT-InSAR). The quantification of soil erosion/deposition was then achieved by integrating horizontal geomorphic changes with interferometric SAR (InSAR)-derived LOS deformation using a weighted least-squares inversion, while the corresponding uncertainty was quantified using the covariance matrix. The proposed image registration method was evaluated using registration errors and deformation uncertainty on stable areas, while the detected soil erosion and deposition were verified based on erosion pin measurements, laser scanning, and manually digitized terrain feature line change directions. The results showed that the proposed registration method achieved subpixel registration accuracy, with a root-mean-square error (RMSE) <5 cm. The derived erosion and deposition were generally comparable with the results obtained by laser scanning and erosion pin monitoring ($R^2 > 0.6$, $p < 0.01$, Nash–Sutcliffe efficiency (NSE) > 0.5), with an RMSE of 3.28 cm according to erosion pin measurements. In addition, the migration of terrain feature lines was found to be generally opposite to the direction of surface horizontal deformation, particularly pronounced at gully heads. This further demonstrated the feasibility of our monitoring results. Overall, this study provided a promising framework for soil erosion monitoring over complex terrain areas based on satellite remote sensing, while further work is still required to improve the accuracy of the method, particularly to improve the capacity of InSAR for monitoring large and rapid changes.**

***Index Terms*—High-resolution optical image, image registration, interferometric synthetic aperture radar (InSAR), Loess Plateau, soil erosion.**

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I. Introduction

TOPOGRAPHICALLY complex areas have been highly susceptible to geomorphic hazards, such as soil erosion [1], landslides [2], and debris flows [3]. Efficient monitoring of soil erosion, particularly over large areas, provides a prerequisite for the implementation of protection strategies and emergency rescue schemes. However, complex topography [4], [5], corresponding to surface heterogeneity and limited accessibility [6], presents great challenges for efficient monitoring [7]. In recent years, remote sensing, with the advantages of multisource data and wide-area coverage, has offered new opportunities for soil erosion monitoring in complex topography. Nevertheless, such studies have primarily been concentrated on small to medium scales, and they have been primarily conducted based on processing 3-D points/digital terrain models (DEMs) produced from laser scanning and/or photogrammetry (e.g., the slice contraction change detection (SCCD) algorithm [8], the adapted DEM of difference (DoD) uncertainty method [9], and the 3-D multiscale model-to-model cloud comparison (3D-M3C2) [10]), while a large-scale soil erosion monitoring on topographically complex areas using spaceborne remote sensing was still scarce.

With the rapid advancement of spaceborne remote sensing technologies, monitoring soil erosion using high-resolution, multitemporal, and multisource satellite imagery (e.g., Sentinel, GaoFen (GF), and WorldView) has gradually become feasible. Various satellite remote sensing techniques have been developed in recent years to enhance the capabilities of monitoring soil erosion, such as interferometric synthetic aperture radar (InSAR) [11], pixel offset tracking (POT) [12], and satellite laser altimetry-assisted DEM differencing [13].

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Among those, InSAR technology offers high precision and all-weather capabilities, which are crucial for surface deformation monitoring at large scales [14], [15]. This makes InSAR well-suited for detecting subtle subsidence or uplift processes caused by geomorphic hazards, such as debris flow [16] and soil erosion [17]. Compared to traditional InSAR technology, multitemporal InSAR (e.g., small baseline subsets InSAR (SBAS-InSAR) [18] and persistent scatterer InSAR (PS-InSAR) [19]) overcomes spatial and temporal decorrelations and atmospheric delay, which significantly improves measurement accuracy and signal-to-noise ratio (SNR) [20]. Meanwhile, the phase optimization algorithm combined with the Stanford method for PS-InSAR (StaMPS-InSAR) identifies denser and more phase-stable points than SBAS-InSAR [21], thereby providing a more robust solution for soil erosion monitoring in the complex terrain.

With the continuous enrichment of remote sensing imagery datasets, POT has also become one of the primary methods for monitoring horizontal geomorphic change [22], particularly in regions with large deformation gradients. The POT method mainly includes synthetic aperture radar (SAR) POT and optical POT [23] and has been widely employed in studies of earthquakes [24], [25], landslides [26], dune migration [27], and glacier flow velocities [28]. Satellite laser altimetry-assisted DEM differencing primarily utilizes high-precision satellite laser elevation data as control points to enhance the accuracy and robustness of DEM differencing [29]. These advancements provide new opportunities for multidimensional monitoring of soil erosion.

Although the above-mentioned satellite remote sensing methods provide powerful tools for monitoring soil erosion in complex terrain, these methods face challenges in practical applications. First, the spatial distribution of satellite laser points (e.g., ICESat-2 and GEDI) is discrete [30] and often fails to fully capture detailed surface changes. Second, soil erosion in the complex terrain often exhibits multidirectional movements of surface materials [31], including both vertical and horizontal deformations. A single-source observation method is incapable of capturing the multidimensional characteristics of soil erosion. Third, InSAR is inherently insensitive to N-S deformation owing to the side-looking observation geometry of SAR [32]. In addition, SAR data from a single orbit were incapable of capturing horizontal deformations, which inevitably limited their application in the complex terrain [33]. Fourth, variations in terrain relief, atmospheric conditions, and imaging geometry often cause geometric inconsistencies among multitemporal remote sensing images [34], [35], which increase the complexity of data processing. Finally, SAR POT is able to provide horizontal deformation in the azimuth and range directions [36], but often fails to directly capture deformation along the N-S or E-W directions.

Given this, the integration of InSAR and optical POT has the potential to provide multisource complementarity and multidimensional observation. InSAR is capable of detecting centimeter- or millimeter-level deformations along the line of sight (LOS), whereas the optical POT enables robust capture of large gradient displacements. In addition, unlike SAR POT, optical POT is able to directly obtain the horizontal displacements. This integration is hypothesized to not only overcome the limitations of single methods regarding directionality and scale but also offer new possibilities for high-accuracy and large-scale soil erosion monitoring in complex terrain. Although previous studies have already combined InSAR and optical offsets for monitoring landslides [37], surface deformation, and geomorphic processes [38], few studies have focused on multidimensional soil erosion monitoring in complex terrain areas (e.g., hilly and gully Loess Plateau).

In addition, the effectiveness of optical POT depends upon the image resolution and the accuracy of image co-registration [39], [40]. The latter has even become more critical, given that high spatial resolution imagery has become increasingly accessible. Several optical image registration approaches have been developed by previous studies [41], [42], [43], [44], such as Harris corner detection [45], [46], scale-invariant feature transform (SIFT) [47], and speeded-up robust features (SURFs) [48]. However, they were often challenged by registration errors caused by geometric distortions within small windows in the complex terrain [49]. In addition, traditional feature point matching methods typically achieve only a pixel-level registration accuracy, which was insufficient for high-precision applications such as soil erosion monitoring that demand subpixel accuracy. The optical flow method proposed by Brigot et al. [50] enables a subpixel registration of multitemporal images. However, this method requires minimal structural differences between image pairs and fails to account for the effect of surface changes, particularly those on topographically complex areas. An enhanced image registration method was therefore required for topographically complex areas, which provided core support for subsequent soil erosion monitoring.

In this study, we aimed to propose an approach for monitoring multidimensional soil erosion based on multisource satellite remote sensing imagery, taking a small catchment with complex topography as a study site. We mitigated geometric distortions in the complex terrain and improved the accuracy of the horizontal change detection by proposing a multimodal high-resolution image subpixel registration method. The complementary observations provided by InSAR and optical POT were jointly utilized to estimate the vertical and slope-normal components of soil erosion, enabling a multidimensional characterization of geomorphic changes. Finally, the proposed registration method was evaluated, and the multidimensional soil erosion and deposition were comprehensively verified. Meanwhile, the advantages and limitations of the proposed method were discussed.

II. Materials and Methods

A. Study Site

Our study was conducted in the first tributary, i.e., Telagou, of the Huangfuchuan catchment [see Fig. 1(b)], located in the hilly and gully Loess Plateau, China. The Huangfuchuan catchment represents a typical topographically complex area, which partly resulted from severe erosion driven by multiple agents, including wind, water, and gravity erosion. Telagou extends across 128.2 km^2 and has elevations ranging from 930

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to 1251 m. The mean annual temperature and precipitation of the catchment are 7.3 °C and 400 mm, respectively. The soil is dominated by the Pisha sandstone, loess, and aeolian sand, which are highly susceptible to water erosion [51], landslides [52], and debris flows [53]. Geomorphic units in this study area are predominantly hillslopes, gully slopes, and gully channels. The terrain of the hillslopes and channels is generally flat (with a slope gradient of 10°–20°), while that of the gully slopes is relatively steep (with a slope gradient of generally over 35° and occasionally over 50°). The catchment experiences severe erosion, primarily including wind erosion, water erosion, and mass movement [54], [55], [56]. Erosion is relatively mild on the hillslopes, whereas it is severe on the gully slopes. Since 1983, vegetation restoration and check dam construction have been implemented in the Telagou catchment to mitigate soil and water loss [57]. Gentle slopes are currently dominated by grassland vegetation, whereas steep slopes still exhibit relatively low vegetation cover. This study focused on a tributary of the Telagou catchment [see Fig. 1(a)], extending over an area of 0.04 km^2. In the study area, an extensive exposure of Pisha sandstone, vegetation degradation, pronounced topographic relief, and intense rainstorms have resulted in severe soil erosion.

The study area experienced stable geological conditions during the study period and was not influenced by major geological events such as fissures or earthquakes. The geological structure is comparatively simple, and no significant geological hazards have been recorded to date. Therefore, geomorphic changes in the study area were considered to primarily result from natural processes, i.e., erosion and deposition.

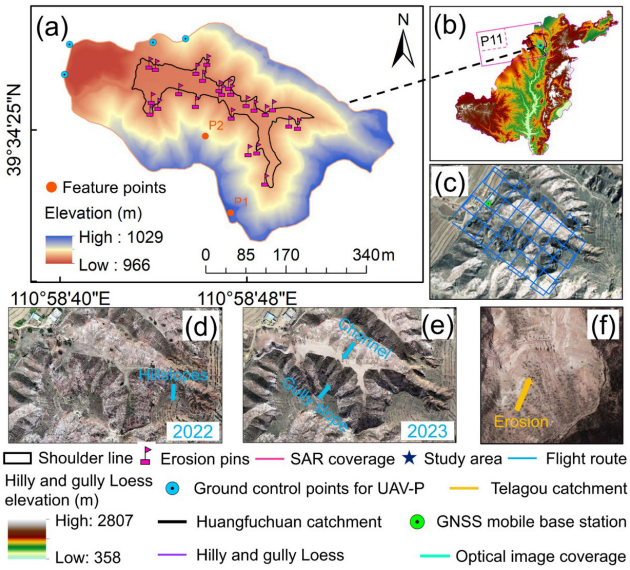


Fig. 1. (a) General information of the study site, including terrain and boundary, with erosion pins, gully shoulder lines and GCPs for UAV-P, (b) location of study site and spatial coverage of high-resolution optical imagery and SAR image, with P11 representing the track number of the SAR image employed in this study, (c) UAV-P flight routesand and RTK installation location, geomorphologic composition of the study area, overlaid on a 0.05 × 0.05 m digital orthophoto model (DOM) derived from oblique photogrammetry [(d) 2022 and (e) 2023], and (f) photographs captured in the study area.

B. Materials Description

1) SAR Datasets: In this study, 19 Sentinel-1A ascending SAR images covering the study area were employed [see Fig. 1(b)], using the Interferometric Wide Swath (IW) imaging mode. Surface displacements in the study area between July 3, 2022 and March 12, 2023 were derived from C-band (5.6 cm) Sentinel-1A satellite images obtained using the terrain observation with progressive scans SAR (TOPSAR) mode. The vertical transmit-vertical receive (VV) dual polarization mode was utilized because it generally provides higher interferometric coherence and was therefore more suitable for deformation monitoring [58]. The SAR images exhibited a mean central incidence angle of 33.66° and a heading angle of 349.5°, with range and azimuth resolutions of 2.33 and 13.99 m, respectively.

2) Data Acquisition by Photogrammetric UAV: The Phantom 4 real-time kinematic UAV platform with DJI FC6310R digital cameras onboard, featuring a focal length of 8.8 mm and an effective resolution of 20 million pixels, was leveraged to acquire very high-resolution imagery of the study catchment on July 29, 2022 and March 16, 2023, respectively. The UAV was operated at approximately 120 m above ground on an intersecting-parallel flight route [see Fig. 1(c)]. The images were captured with forward and side overlap of 80% and 70%, respectively, and the tilt angle of the camera was set to −45°. Four ground control points (GCPs) were established and surveyed in the Telagou catchment at the outlet of the gully channel and in areas that were easily accessible for deployment and observation, with coordinates referenced to the CGCS2000 [see Fig. 1(a)]. The flights were carried out during favorable weather conditions to minimize atmospheric interference in the aerial imagery. Aerial triangulation of the acquired photographs was performed using ContextCapture Center 10.2, followed by the generation of uncrewed aerial vehicle photogrammetry (UAV-P) point clouds and the digital orthophoto images (DOMs) with a spatial resolution of 0.05 m.

3) High-Resolution Optical Imagery: The temporal alignment of satellite imagery (e.g., SAR and optical imagery) with field surveys is often constrained by imaging schedules and satellite revisit intervals. Given this, satellite images acquired on dates closest to those of field survey dates were selected to minimize temporal discrepancies and to provide a basis for subsequent validation.

GF-7 is an optical Earth observation satellite with high spatial resolution, launched by China National Space Administration (CNSA) and China Aerospace Science and Technology Corporation (CASC) on November 3, 2019, with a designed lifespan of eight years. The GF-7 satellite carries a dual-line-array stereo camera and a laser altimeter. The stereo camera acquires panchromatic imagery at a 0.65-m spatial resolution and multispectral imagery at a 2.6-m resolution. The GF-7 satellite imagery was obtained on July 16, 2022 and March 14, 2023 from China Centre for Resources Satellite Data and Application (CRESDA) (https://data.cresda.cn/#/home) and was employed to assess the reliability of the proposed registration method.

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The Jilin-1 Kuanfu-01B (JL1KF01B) is a high-resolution remote sensing satellite launched by Chang Guang Satellite Technology Company Ltd. on September 15, 2020. The JL1KF01B satellite is equipped with a high-performance panchromatic and multispectral camera, which provides satellite imagery at a panchromatic resolution of 0.5 m and a multispectral resolution of 2.0 m. These data are publicly accessible to educational and academic users upon application. The JL1KF01B data that covered the study area were acquired on July 24, 2022 and March 16, 2023 from the Jilin-1 platform (https://www.jl1mall.com/), with the imaging mode set to wide-swath and a coverage area of 150 × 150 km. The JL1KF01B satellite imagery was utilized not only to evaluate the reliability of the proposed registration method but also to implement the POT analysis.

4) Field Erosion Pin Measurement Data: Geomorphic changes resulting from soil erosion were monitored using field erosion pins, with each pin being 1.2 m long. A total of 60 field surveys were carried out along the mainstream channel of the study catchment. Due to the steep terrain and low accessibility within the study catchment, erosion pins were predominantly distributed across flat areas of less than 10° (e.g., gully channels). The field survey boundaries and erosion pin coordinates were measured using the global navigation satellite system real-time kinematic (GNSS-RTK), adopting the same coordinate system as the UAV-P point clouds. Erosion pins were manually installed prior to the first UAV-P data collection (July 29, 2022). The second UAV-P data was acquired, and the erosion situation was concurrently measured using the pins (March 16, 2023) after the erosion and deposition phenomena occurred. However, the erosion pins at 20 field survey sites were destroyed by storm-generated runoff, while the remaining 40 field survey sites were used for the analysis.

C. Methodology

Our study was conducted through the following four steps (see Fig. 2). First, the LOS deformation in the Telagou catchment was obtained using a phase optimization algorithm in conjunction with the StaMPS-InSAR technology. Second, multitemporal high-resolution remote sensing imagery were acquired, and the multiscale histogram and iterative optical flow (MSH-IOF) method was proposed to achieve subpixel image registration in the complex terrain. Third, panchromatic and multispectral bands were fused using the contour-aware dynamic low-high frequency integration (CDFInet) algorithm, followed by dimensionality reduction through the weighted principal component analysis (WPCA) method. Thereby, 1-D horizontal displacements were derived using the optical POT. The deformations derived from InSAR and optical POT were combined to obtain the multidimensional soil erosion. Finally, the proposed registration method and the derived soil erosion were separately evaluated based on LiDAR-based high-accuracy monitoring.

1) Methods for Data Processing:

a) Registration of high-resolution remote sensing image for complex terrain: Image registration was performed for GF-7 and JL1KF01B satellite imagery. However, due to the complex terrain and diverse surface textures in the hilly and gully region, the preliminary image registration was suboptimal, with local registration residuals exceeding 3 pixels. We proposed a multimodal high-resolution remote sensing image alignment method called MSH-IOF to further improve the quality of the preliminary image registration, as shown in Fig. 3. MSH-IOF combined the multiscale histogram of local main orientation (MS-HLMO) feature matching method with iterative optical flow to progressively reduce geometric discrepancies between images. The MS-HLMO enabled the registration of optical images acquired at different resolutions and times through extracting and matching feature points with rotation and scale invariance [59], [60], [61], making it well suited for complex hilly and gully terrains with varying slopes, aspects, and surface textures. However, MS-HLMO typically achieved pixel-level registration accuracy, which did not satisfy the subpixel accuracy requirements of high-precision applications (e.g., soil erosion monitoring). Optical flow estimated the motion of pixels across image sequences

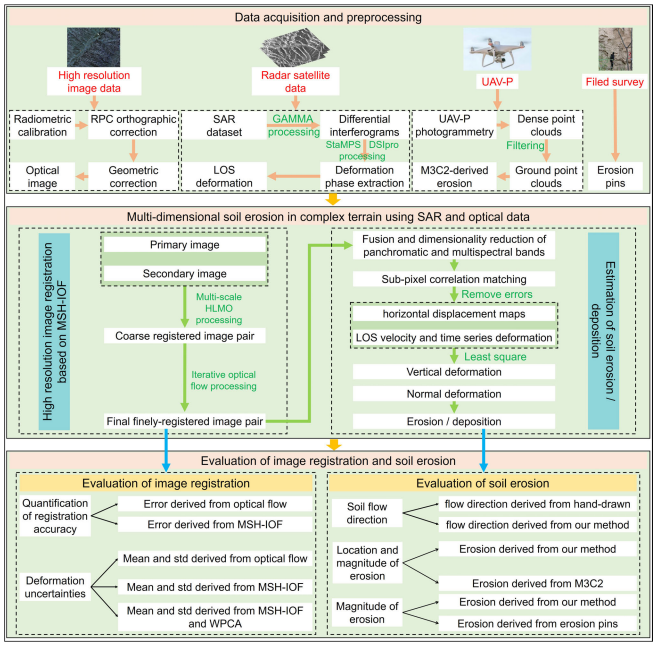


Fig. 2. Overall workflow of this study.

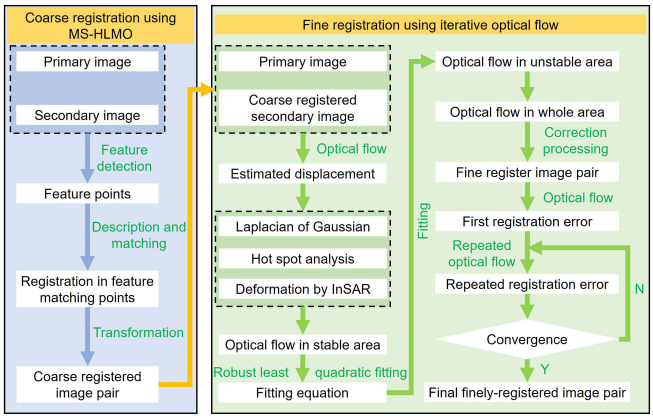


Fig. 3. Overall flowchart of the MSH-IOF method.

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and provided subpixel accuracy for image alignment [62], [63]. However, this method requires minimal structural differences between image pairs and fails to account for the effect of surface changes. To address these limitations, optical images were coarsely registered using MS-HLMO. Building upon the registration results of MS-HLMO, the optical flow (GeFolki) algorithm proposed by Brigot et al. [50] was employed to refine the registration results and achieve subpixel alignment accuracy. The initial optical flow between the primary and secondary images was estimated and directly applied to stable areas, which were automatically identified as regions exhibiting no obvious surface changes. The optical flow parameters for the deformation regions were then estimated by performing robust least-squares fitting based on the optical flow parameters of the stable area. Finally, an iterative strategy grounded in the optical flow was implemented to incrementally minimize residual geometric discrepancies between the images, which ultimately achieved subpixel registration accuracy.

A primary image (A) and a secondary image (B) were considered during the registration. For a pixel (x, y) in the image, it was assumed that pixel intensities remained constant from one frame to the next, which can be written as follows:

$$f(x + dx, y + dy, t + dt) = f(x, y, t) \tag{1}$$

where (dx, dy) represents the displacement vector and f denotes the pixel intensity. Equation (1) is expanded using the Taylor series, as shown in the following equation:

$$f(x + dx, y + dy, t + dt) = f(x, y, t) + \partial f/\partial x dx + \partial f/\partial y dy + \partial f/\partial t dt + \omega \tag{2}$$

where ω represents the higher order infinitesimal term in the Taylor series and can be neglected. The optical flow constraint equation was derived by combining (1) and (2), as shown in the following equation:

$$f_x u + f_y v + f_t = 0 \tag{3}$$

where $f_x = \partial f/\partial x$, $f_y = \partial f/\partial y$, and $f_t = \partial f/\partial t$ represent the gradients of image intensity in the x-, y-, and t-directions, respectively. $u = dx/dt$ and $v = dy/dt$ denote the horizontal and vertical components of the optical flow, respectively.

The GeFolki algorithm features strong robustness to intensity variations [64], [65], thereby mitigating the effects of local illumination differences caused by complex terrain on image registration. Accordingly, the GeFolki algorithm was employed to further improve the registration accuracy. First, the initial optical flow between the primary and secondary images was calculated using the GeFolki method. Areas containing noncorresponding features between the two images produced anomalous optical flow vectors, characterized by magnitudes and directions that differed significantly from those of neighboring pixels. The Laplacian of Gaussian (LoG) was then used to identify these anomalous optical flow vectors, while hotspot analysis of the deformation derived using InSAR was performed to detect additional deformed regions. Areas exhibiting both anomalous optical flow patterns and significant deformation signals were classified as deforming areas, whereas the remaining areas were regarded as stable areas. The identified stable areas did not only include relatively flat regions (e.g., buildings and roads) but also areas characterized by different slope conditions and geomorphic units (e.g., gully slopes and hillslopes). Second, the optical flow parameters for the deforming and stable areas were retrieved. The initial optical flow parameters were directly applied to the stable areas. The optical flow parameters for the deformed region were derived through applying robust least squares fitting to the parameters of stable areas. Third, the optical flow parameters for the stable and deforming areas were combined to register the secondary image. Finally, the optical flow method was iteratively updated to further optimize the registration.

A quadratic polynomial surface fitting equation was used to perform robust least squares fitting on the initial optical flow of the stable area, as shown in the following equation:

$$z_n = a_0 + a_1 x_n + a_2 y_n + a_3 x_n y_n + a_4 x_n^2 + a_5 y_n^2 \tag{4}$$

where z_n represents the initial optical flow of pixel n on the stable area and n is a positive integer. x_n and y_n denote the coordinates of pixel n, while a_0, a_1, a_2, a_3, a_4, and a_5 are the coefficients to be determined.

To estimate the above coefficients, the matrix form of the error equation was derived within the framework of condition adjustment based on the relationship between the observed data (i.e., z_n) and the fitting model (i.e., (4), as shown in the following equations:

$$V = MK - Z \tag{5}$$

$$V = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix};\ K = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix};\ Z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \tag{6}$$

$$M = \begin{bmatrix} 1 & x_1 & y_1 & x_1 y_1 & x_1^2 & y_1^2 \\ 1 & x_2 & y_2 & x_2 y_2 & x_2^2 & y_2^2 \\ \vdots & \vdots & & \vdots\ \vdots\ \vdots & & \\ 1 & x_n & y_n & x_n y_n & x_n^2 & y_n^2 \end{bmatrix} \tag{7}$$

where V represents the residual vector of the observations, K is the vector of unknown coefficients, M denotes the design matrix, and Z is the observation vector composed of the initial optical flow on the stable area. K was calculated using the following equation based on the principle of least squares:

$$K = \left(M^T P M\right)^{-1} M^T P Z \tag{8}$$

The weight matrix was calculated through the Institute of Geodesy and Geophysics (IGGs)-III method to mitigate the influence of outliers [66], as shown in the following equations:

$$\bar{P}_{ij} = \begin{cases} P_i, & |\frac{V_j}{\sigma}| < K_0 \\ P_{ij} \dfrac{K_0}{|\frac{V_j}{\sigma}|} \left(\dfrac{K_1 - |\frac{V_j}{\sigma}|}{K_1 - K_0} \right)^2, & K_0 \le |\frac{V_j}{\sigma}| \le K_1 \\ 0, & K_1 < |\frac{V_j}{\sigma}| \end{cases} \tag{9}$$

$$\sigma = \sqrt{\frac{V^T P V}{c - t}} \tag{10}$$

where σ represents the standard deviation of unit weight, c is the number of observations, and t indicates the minimum

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number of observations. To balance robustness and estimation efficiency, K_0 and K_1 were set to 1.5 and 3, respectively, with reference to Yang [67] and Zhang et al. [68].

The optical flow changes between the corrected image and the primary image in the stable area were theoretically zero. Accordingly, the root-mean-square error (RMSE) of the optical flow changes in the stable area was calculated and employed as a metric for evaluating registration accuracy, as shown in the following equation:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(W_x^2 + W_y^2\right)} \tag{11}$$

where W_x and W_y represent the optical flow of pixels in the stable area in the vertical and horizontal directions, respectively. The optical flow method was iteratively updated based on the initial optical flow registration, while RMSE was used to evaluate registration accuracy. The optical flow method was terminated when the difference between the RMSE of the previous and next layers decreased monotonically, and its absolute value was less than ε, as shown in the following equation:

$$|RMSE_{m-1} - RMSE_m| < \varepsilon. \tag{12}$$

The reliability of the proposed registration method was evaluated using multitemporal GF-7 and JL1KF01B satellite images, with registration errors in stable regions used as an accuracy metric for the registration. After achieving registration work, the high-resolution panchromatic band was fused with the spectrally rich multispectral bands to simultaneously enhance spatial resolution and spectral fidelity of the remote sensing imagery. This provided a superior quality input for subsequent processes of deformation extraction, thereby improving the applicability and robustness of the POT method in the complex terrain. In this study, a free open-source CDFInet software developed by Yi et al. [69] was adopted to fuse the panchromatic and multispectral bands of high-resolution remote sensing imagery. The CDFInet was used to perform multidirectional decomposition on the image by introducing contour wavelet transformation, which effectively captures fine details. However, the POT method primarily relies on single-channel data for matching and fails to directly process the fused high-dimensional image data. Therefore, the dimension reduction was performed on the fused imagery using the WPCA method to facilitate subsequent change detection using the POT method [70].

b) SAR data preprocessing: We obtained the LOS deformation of the study catchment by using GAMMA [71] and DSIpro [72] in combination with StaMPS-InSAR [73]. First, the image acquired on November 12, 2022 was selected as the primary image and served as the reference for registering the remaining images. We utilized the 30-m resolution Copernicus DEM (COP-DEM) derived by European Space Agency to perform georeferencing and correct terrain-induced phase errors. Temporal and spatial decorrelation were mitigated by setting the time baseline to 150 days and the perpendicular baseline to 200 m. The amplitude dispersion threshold (less than 0.5) was used [17], and the phase noise standard deviation (γ) (less than 1.2) was calculated for each pixel pair, including the pixel [74] to increase the quantity and quality of PS points. The LOS deformation was achieved based on a reference point on stable areas, through performing 3-D phase unwrapping [73], and correcting the atmospheric delay of the interferograms using the generic atmospheric correction online service (GACOS) [75], [76], [77]. The LOS deformation result was positive when the ground objects moved closer to the satellite. Conversely, the deformation result was negative when the ground object moved away from the satellite.

TABLE I
QUANTIFICATION OF REGISTRATION ACCURACY FOR DIFFERENT DATASETS BY THREE METHODS

Data	MS-HLMO (m)	Optical flow (cm)	MSH-IOF (cm)
GF-7	0.9	13.5	4.1
JL1KF01B	0.7	30.7	2.7

c) Estimation of the horizontal deformation using optical pixel offset-tracking: The horizontal displacement was obtained from optical images using the POT method. Optical POT estimates surface displacement by tracking the relative movement of pixels between two co-registered images using the phase correlation method [78]. The method employs a two-step iterative process with variable window sizes to improve both correlation robustness and displacement accuracy. The initial window size is chosen to maximize the correlation between the two images, while the final window size enables a more precise estimation of ground deformation [23]. The accuracy of the horizontal deformation derived from optical POT is largely determined by the spatial resolution of the optical imagery [79]. Given that the spatial resolution of the JL1KF01B satellite imagery is superior to that of GF-7, this study employed JL1KF01B satellite imagery for the implementation of POT. JL1KF01B satellite imagery provides a 0.5-m resolution for the panchromatic band and 2-m resolution for the multispectral bands (i.e., blue, green, red, and infrared bands). We used the fused band for correlations after evaluating the performance of the panchromatic and fused bands. The POT method involves five key parameters, which are the initial window size (i), final window size (f), robustness iteration (r), sampling step (s), and SNR threshold [80]. We gradually increased the initial values of each parameter from the lower limit of the allowed range in fixed steps until the upper limit was reached, and compared the deformation standard deviation in stable areas to obtain the optimal parameter values. To obtain higher quality horizontal deformation, i, f, r, and s were set to 8, 8, 2, and 1, respectively [81]. To obtain reliable horizontal deformation, an SNR threshold of 0.95 and the mean value and standard error of the N-S and E-W deformations were employed to remove potential outliers [82]. Image detrending and nonlocal filtering were then conducted to further remove noise from the initial correlation results [83]. The measured deformation in the E-W or N-S directions was positive when the ground objects moved toward the east or north, respectively. The uncertainties of deformation assessed by high-resolution imagery using different methods (e.g.,

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optical flow, MSH-IOF, and MSH-IOF + WPCA) were evaluated through the standard deviation of the deformation observed on stable areas. In addition, the magnitude and direction of surface horizontal deformation were calculated based on the N-S and E-W displacements obtained from optical POT [84], as shown in the following equations:

$$d = \sqrt{d_{EW}^2 + d_{NS}^2} \tag{13}$$

$$\delta = \begin{cases} 90 - \frac{180}{\pi}\left|\tan^{-1}\left(\frac{d_{EW}}{d_{NS}}\right)\right|, & d_{EW} > 0, d_{NS} > 0 \\ 180 - \frac{180}{\pi}\left|\tan^{-1}\left(\frac{d_{EW}}{d_{NS}}\right)\right|, & d_{EW} > 0, d_{NS} < 0 \\ 180 + \frac{180}{\pi}\left|tan^{-1}\left(\frac{d_{EW}}{d_{NS}}\right)\right|, & d_{EW} < 0, d_{NS} < 0 \\ 360 - \frac{180}{\pi}\left|tan^{-1}\left(\frac{d_{EW}}{d_{NS}}\right)\right|, & d_{EW} < 0, d_{NS} > 0 \end{cases} \tag{14}$$

where d is the magnitude of surface horizontal deformation, and δ is the direction of surface horizontal deformation. d_{EW} and d_{NS} are the E-W and N-S deformations derived from optical POT, respectively.

d) Estimation of geomorphic changes (erosion/deposition) by combining optical POT and MT-InSAR: The multidimensional changes induced by soil erosion were inverted based on the LOS and horizontal displacements. Because the time span of the horizontal deformation (July 24, 2022 to March 16, 2023) did not fully coincide with that of the multitemporal interferometric SAR (MT-InSAR) results (July 3, 2022 to March 12, 2023), the LOS deformation corresponding to the period of horizontal deformation was extracted from the cumulative MT-InSAR time-series deformation. In addition, the InSAR-derived deformation was adopted as the reference framework to minimize the effects of differences in spatial scale, and the high-resolution observations obtained from optical POT were spatially aggregated to ensure the spatial support was consistent with the InSAR-derived deformation. Specifically, the optical POT observations falling within each InSAR pixel were averaged to derive representative values, allowing the different datasets to be compared under a common ground sampling distance. Accordingly, all comparisons were performed at the original spatial resolution of the InSAR-derived deformation (approximately 14 m). Considering the side-looking geometry of Sentinel-1 data, a 3-D deformation observation equation was constructed by combining the LOS, E-W, and N-S deformations within the same time period to obtain the vertical deformation, as shown in (15). Negative vertical deformation values indicate subsidence, whereas positive values indicate uplift

$$\begin{bmatrix} d_{los} \\ d_{EW} \\ d_{NS} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \cdot \cos\varphi & \sin\theta \cdot \sin\varphi \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} d_u \\ dew \\ dns \end{bmatrix} \tag{15}$$

where θ and φ denote the mean incidence and heading angles of SAR satellites, respectively. d_{los} represents the change in the LOS direction. d_u, dew, and dns indicate the actual 3-D deformation.

Since soil erosion involves the degradation of soil layer thickness, its manifestations vary across different terrains. Soil erosion primarily manifests as cumulative vertical deformation on flat areas [85], while it appears as cumulative normal deformation on steep regions. Therefore, the erosion/deposition process can be regarded as a combined effect of vertical deformation in flat areas and normal deformation on slopes. The results in the slope-normal direction were obtained using the method described by Fan et al. [86], as shown in (16) and (17). Negative slope-normal deformation values corresponded to erosion of the gully slopes, whereas positive values indicated deposition

$$\begin{bmatrix} d_u \\ d_{ns} \\ d_{ew} \end{bmatrix} = \begin{bmatrix} -\sin\alpha & \cos\alpha \\ \cos\alpha \cdot \cos\beta & \sin\alpha \cdot \cos\beta \\ \cos\alpha \cdot \sin\beta & \sin\alpha \cdot \sin\beta \end{bmatrix} \cdot \begin{bmatrix} d_{slope} \\ d_{normal} \end{bmatrix} \tag{16}$$

$$W = \left(Q^T P Q\right)^{-1} Q^T P L \tag{17}$$

where α denotes the slope gradient, and β is the slope aspect. d_{slope} and d_{normal} indicate the changes along the slope-aspect and the slope-normal direction, respectively. $[d_u\, d_{ns}\, d_{ew}]^{-1}$ and Q denote the observation vector and coefficient matrix in (16). $W = [d_{slope}\, d_{normal}]^{-1}$ denotes the displacement vector to be determined. P is the weight matrix and was obtained from the standard deviation in the stable area to reduce the differences in accuracy achieved by different techniques [23]. The observation covariance matrix incorporated the uncertainties of the input deformation measurements, including the InSAR-derived LOS deformation and the optical POT-derived horizontal deformation. These uncertainties were propagated to the final erosion/deposition estimates through the least-squares adjustment. The uncertainty associated with the estimated erosion/deposition was then quantified using an analytical error propagation, as shown in the following equations:

$$\sigma_0 = \sqrt{V^T P V/(n-m)} \tag{18}$$

$$\sigma^2 = \sigma_0^2 \left(Q^T P Q\right)^{-1} \tag{19}$$

where σ_0 is the unit weight standard deviation, V represents the residual vector of (16), n denotes the number of observations, and m denotes the number of necessary observations.

2) Verification Approaches: To assess the credibility of the estimated erosion/deposition, we compared the erosion/deposition with M3C2 and field erosion pin measurements, respectively. The M3C2 and erosion pin observations covered the period from July 29, 2022 to March 16, 2023, while the estimated erosion/deposition corresponded to the period from July 24, 2022 to March 16, 2023. Owing to satellite acquisition schedules and revisit intervals, exact temporal alignment between the satellite imagery and the field surveys was not achieved. Consequently, the remaining temporal mismatch may introduce additional uncertainty into the comparison. Nevertheless, the selected datasets represented the closest temporal correspondence achievable in this study, thereby minimizing the influence of temporal inconsistency on the validation results. The accuracies of the estimated erosion/deposition were evaluated using the coefficient of

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determination (R^2), RMSE, and Nash–Sutcliffe efficiency (NSE).

a) Terrain change (erosion/deposition) detection by M3C2: To reduce the registration errors caused by human and systematic biases, co-registration among UAV-P point cloud datasets was performed using the iterative closest point (ICP) algorithm [87]. The buildings within the catchment were used as stable areas for co-registration. The co-registration was done in CloudCompare version 2.10 software (https://www.danielgm.net/cc), where the target registration error was less than 0.02 m. The cloth simulation filter (CSF) methods [88], [89], also implemented in CloudCompare, were then applied to filter the UAV-P point clouds [90]. This method involves four key parameters of the scene (s), cloth resolution (r), maximum number of iterations (i), and classification threshold (c). Following previous application on the Loess Plateau [91], the parameters of s, r, i, and c were set to steep slope and slope processing, 2 m, 500, and 0.5 in our study, respectively. Nevertheless, the method was prone to misclassifying ground points as nonground on steep areas, such as gullies [92]. Given this, the CSF point clouds were then manually inspected and adjusted in Terrasolid 2016 to ensure classification accuracy.

The M3C2 algorithm enables the detection of topographic changes in topographically complex regions via direct comparison of point clouds [93], [94], without the need for point cloud interpolation. The algorithm is characterized by two primary parameters: the normal scale (D) and projection scale (d). The M3C2 algorithm, implemented in CloudCompare v2.10, was employed to calculate the distance between the two point clouds. Referring to a previous implementation on the Loess Plateau [91], D and d were set to 0.4 and 0.5 in this study, respectively.

The M3C2-derived changes were used to validate the reliability of the slope-normal deformation derived by our method using the metrics of R^2 and NSE. Because the datasets used had different spatial resolutions, direct pixel-to-pixel or point-to-point comparisons were inappropriate. Therefore, we performed a spatial aggregation on the soil erosion/deposition observations obtained from M3C2 to ensure that the spatial support was consistent with the InSAR-derived deformation. In total, 150 comparison samples were used for the validation.

b) Erosion/deposition measurements using erosion pin: Because valid PS observations were unavailable near some erosion pins due to insufficient interferometric coherence, only field sites with corresponding PS observations were selected for validation. Consequently, a total of 24 field survey sites were included in the validation [see Fig. 1(a)].

The erosion and deposition derived using our method were compared with the mean values measured by erosion pins within each survey area. The accuracy of the vertical deformation for July 2022 to March 2023 was then evaluated against erosion pin measurements using R^2, NSE, and RMSE, as shown in the following equations:

$$R^2 = \frac{\left[\Sigma_{i=1}^{N}\left(M_i - \overline{M}_i\right)\cdot\left(U_i - \overline{U}_i\right)\right]^2}{\left[\Sigma_{i=1}^{N}\left(M_i - \overline{M}_i\right)^2 \cdot \Sigma_{i=1}^{N}\left(U_i - \overline{U}_i\right)^2\right]} \tag{20}$$

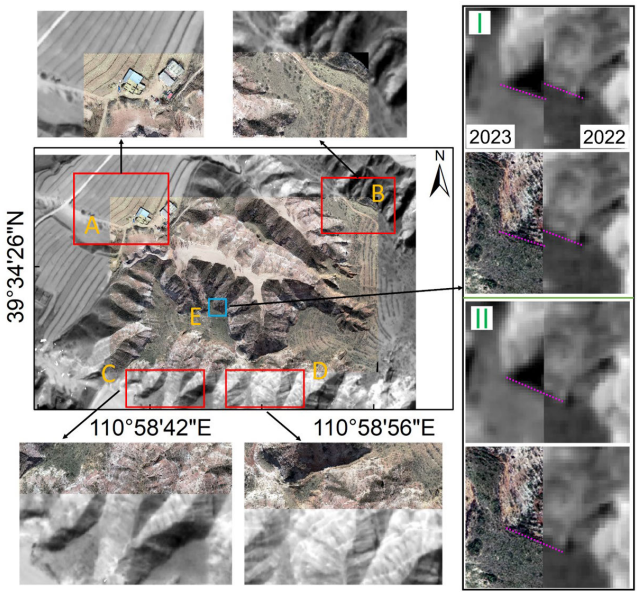


Fig. 4. Registration results across different geomorphic types.

TABLE II
REGISTRATION ACCURACY STATISTICS ACROSS SLOPE CLASSES AND GEOMORPHIC TYPES

Geomorphic types (slope classes)	Registration error (cm)	
	GF-7	JL1KF01B
Gully slopes (>30°)	4.6	3.1
Gully slopes (20°-30°)	4.2	2.9
Hillslopes (10°-20°)	3.3	2.3
Building and road (<10°)	2.6	2.0

$$NSE = 1 - \frac{\Sigma_{i=1}^{N}\left(M_i - U_i\right)^2}{\Sigma_{i=1}^{N}\left(M_i - \overline{M}_i\right)^2} \tag{21}$$

$$RMSE = \sqrt{\Sigma_{i=1}^{N}\left(U_i - M_i\right)^2 / N} \tag{22}$$

where N is the number of samples. M_i represents erosion and deposition elevations measured by erosion pins, and $\overline{M}_i$ is the average of observed values. U_i denotes the vertical deformation obtained in this study, and $\overline{U}_i$ is the average of U_i.

III. RESULTS

A. Evaluation of Image Registration

1) Registration Accuracy of MSH-IOF: The JL1KF01B and GF-7 satellite images, which were co-registered using the MS-HLMO feature-point matching approach, achieved a registration error of less than 1 m (within 1–2 pixels) (see Table I). On top of the initial feature-point matching, the optical flow algorithm achieved a subpixel registration accuracy (i.e., 30.7 cm for JL1KF01B and 13.5 cm for GF-7). After the implementation of the proposed MSH-IOF method, the registration error was further reduced, with that of JL1KF01B and GF-7 data being 2.7 and 4.1 cm, respectively.

We selected flat areas (see A and B of Fig. 4), gully slopes (see C and D of Fig. 4), and gully head (see E of Fig. 4) to verify the registration results. On flat areas, the images registered

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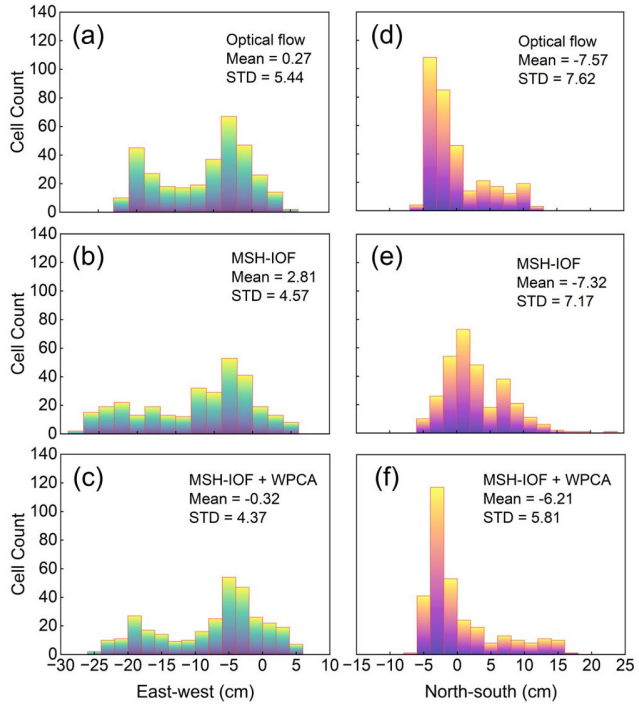


Fig. 5. Histogram of deformation in (a)–(c) E-W direction and (d)–(f) N-S direction using different methods: (a) and (d) optical flow, (b) and (e) MSH-IOF, and (c) and (f) MSH-IOF combined with WPCA.

using the proposed method showed good agreement with the DOM image in fine details such as road networks and building boundaries. On gully slopes, the terrain feature lines were well aligned with the corresponding structures in the DOM image. In contrast, the conventional registration method still exhibited noticeable misalignment at the gully head, though it achieved satisfactory registration on the flat areas (see region I of Fig. 4), highlighting the greater challenge of image registration in this geometrically complex area. Although gully heads are generally characterized by active soil erosion, visual inspection of the DOM imagery and the surrounding terrain revealed no large-scale geomorphic changes that could account for the observed registration misalignment. Therefore, the observed misalignment was considered more likely to result from registration errors than from actual erosion. After registration using the proposed method, the terrain feature lines in this area were well aligned with the corresponding structures in the DOM image, with no obvious misalignment observed (see region II of Fig. 4). In addition, the registration error gradually increased with an increasing terrain slope gradient (see Table II). The largest errors were observed on gully slopes, whereas the smallest errors were found on building and road areas.

2) Uncertainty of Deformation Derived Based on the Registered High-Resolution Imagery: The average and standard deviations of deformation obtained using the optical flow method were 0.3 and 5.4 cm in the E-W direction, and −7.6 and 7.6 cm in the N-S direction [see Fig. 5(a) and (d)]. The mean and standard deviation of E-W deformation obtained using the MSH-IOF were 2.8 and 4.6 cm, respectively [see Fig. 5(b)], while those for N-S deformation obtained using the MSH-IOF method were −7.3 and 7.2 cm, respectively [see Fig. 5(e)]. The differences in the mean and standard deviation of N-S deformation derived by the MSH-IOF and the optical flow methods were relatively small [see Fig. 5(d) and (e)], whereas those for the standard deviation of E-W deformation were more noticeable [see Fig. 5(a) and (b)]. Specifically, the MSH-IOF method reduced the uncertainty (quantified by the standard deviation of deformation results) by 16% in the E-W direction and 6% in the N-S direction as compared to the optical flow method.

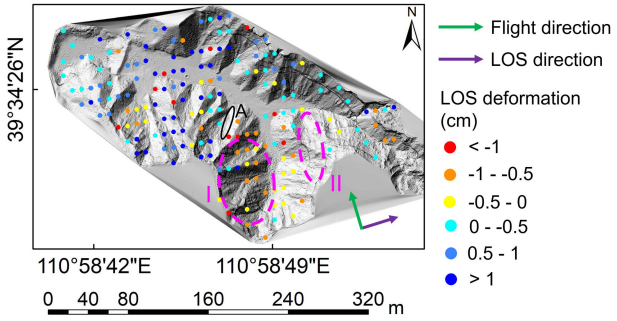


Fig. 6. LOS deformation derived by InSAR.

Compared to the MSH-IOF method, the fusion of panchromatic and multispectral imagery resulted in reducing the uncertainty by 4.38% in the E-W direction and 19% in the N-S direction, respectively. The deformation associated with fused imagery exhibited the lowest uncertainty and the highest monitoring capability. After implementing all procedures embedded in the proposed method, the original uncertainties in both E-W and N-S deformation were reduced by 19.7% and 23.7%, respectively.

B. Evaluation of Soil Erosion Monitoring for Complex Terrain

1) Spatial Characteristics of Soil Erosion:

a) Changes in the LOS direction: The LOS deformation exhibited pronounced spatial heterogeneity (see Fig. 6). During the study period, the majority of the detected LOS displacements were found in the southeastern part of the catchment and particularly on gully slopes, reaching a maximum of −1.8 cm. The deformation was primarily distributed along gully heads and gully slopes, whereas deposition mainly occurred in the middle and bottom of the gully slopes, resulting in a prominent LOS uplift up to 2.1 cm in the corresponding area. LOS deformation was predominantly observed on steep slopes, whereas flatter areas exhibited relatively smaller deformations. A further comparison between slope aspect and radar LOS direction revealed that deformation signals were more pronounced in areas in which the main movement direction of the slope was closely aligned with the radar observation direction (e.g., the slope corresponding to I in Fig. 6). Notably, the laser scanning results detected significant erosion/deposition in area II in Fig. 6, whereas InSAR did not detect significant LOS deformation signals in these regions.

The time-series LOS deformation [see Fig. 7(a) and (b)] showed that points P1 and P2 [see Fig. 1(a)], located at the

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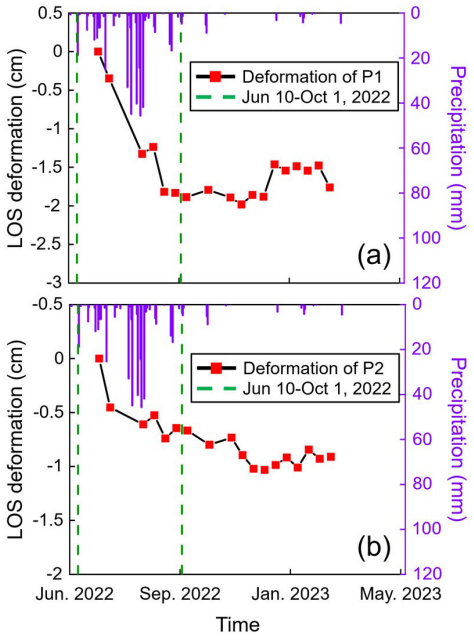


Fig. 7. Time-series LOS deformation of the two selected points (a) P1 and (b) P2.

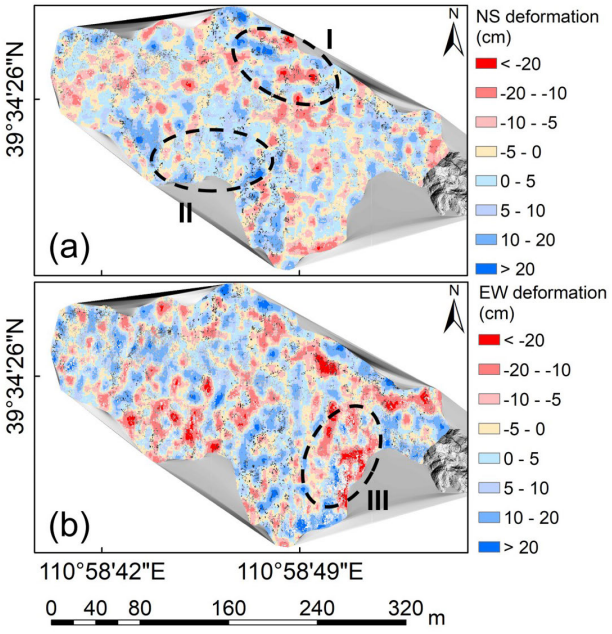


Fig. 8. (a) N-S displacement and (b) E-W displacement derived from optical imagery.

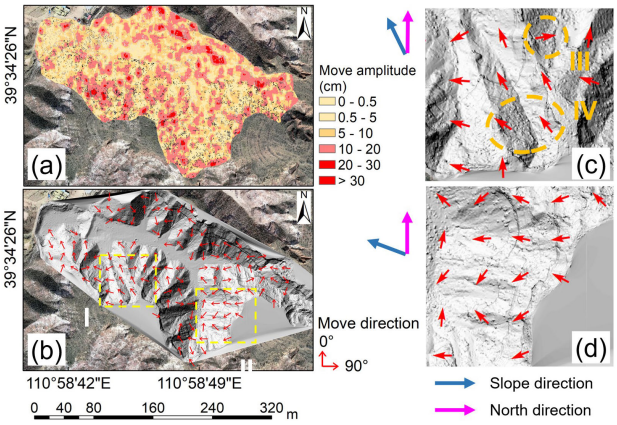


Fig. 9. (a) Magnitude and (b) direction of surface horizontal deformation obtained from the joint inversion of the N-S [see Fig. 8(a)] and E-W [see Fig. 8(b)] deformation components, and (c) and (d) enlarged views of the movement directions within the regions marked by dotted boxes in (b).

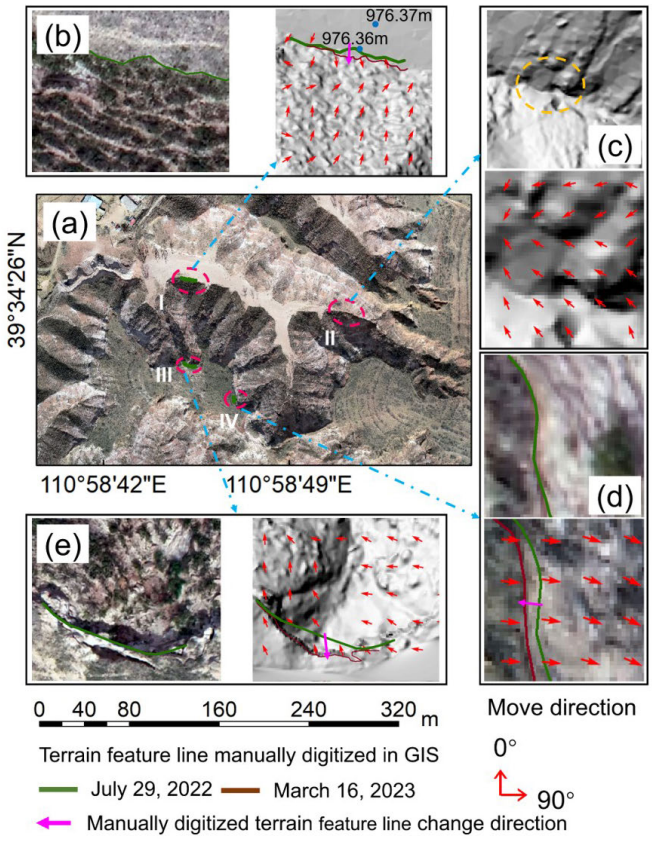


Fig. 10. (a)–(e) Comparison of changes in directions of manually digitized terrain feature lines and observed directions of surface horizontal deformations. The background images are DOMs acquired on July 29, 2022 and March 16, 2023. Lines in different colors represented the positions of terrain outlines at different points in time. Red arrows indicated the direction of surface horizontal deformations derived from optical imagery, while purple arrows represented the directions of manually digitized terrain feature line changes.

gully head and gully slope, experienced persistent negative deformation throughout the monitoring period, with cumulative deformations being approximately −1.7 and −1.1 cm, respectively. Both points exhibited a continuous accumulation of negative deformation between July and October 2022, which was generally coincident with the frequent precipitation events during the same period. Subsequently, the rate of negative deformation accumulation slowed down.

b) Changes in the horizontal direction: Surface movements also exhibited pronounced regional differentiation in both the N-S and E-W directions (see Fig. 8). Deformation was mainly distributed in areas with steep slopes and gully heads. The maximum NS deformation was approximately −30.1 cm, while the maximum EW deformation reached

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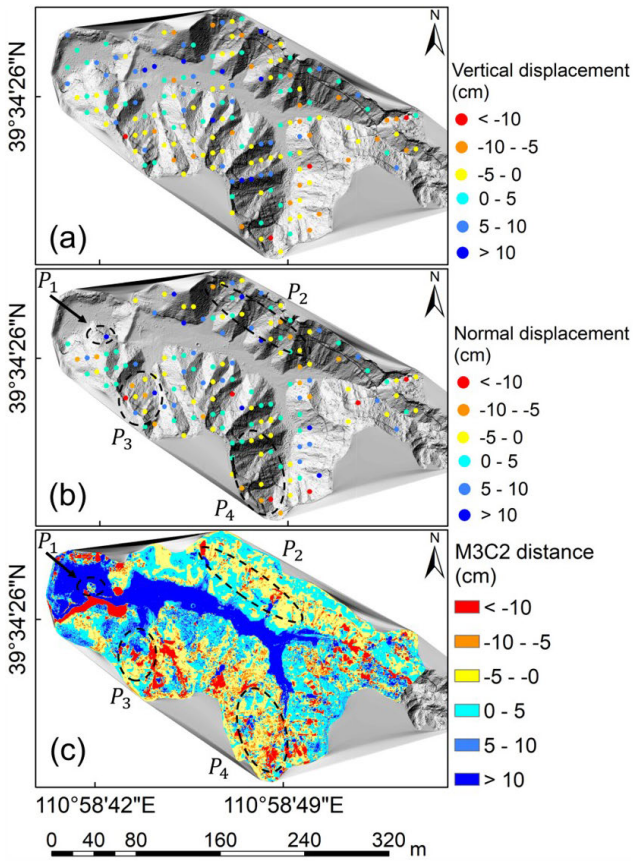


Fig. 11. (a) and (b) Vertical and normal results obtained using the proposed method and (c) normal deformation derived by the M3C2 algorithm.

approximately −38.2 cm, occurring at the gully head of II in Fig. 8(b). The NS displacement revealed significant downslope movement and material accumulation in the gully heads and gully slopes of regions I and II [see Fig. 8(a)], whereas evident EW displacement was concentrated along the gully head and gully slope in region III [see Fig. 8(b)]. These findings were consistent with the regions of significant topographic change identified by the M3C2 method.

Fig. 9(a) clearly illustrates the highly dynamic activity (greater than 20 cm) at the gully heads and relatively slow deformation (less than 10 cm) on the gully slopes. The overall movement within the study catchment was relatively stable (less than 10 cm), with an average displacement of 8.5 cm. The maximum horizontal displacement was 32.8 cm (approximately 65 pixels). The relationship between the direction of surface horizontal deformation and the slope aspect varied across different regions [see Fig. 9(b)]. On some steep slope areas, the two directions were generally consistent [see Fig. 9(c) and (d)], with surface deformation migrating downslope along the aspect. In contrast, in areas such as the middle of gully slopes, the direction of surface horizontal deformation deviated to some extent from the slope aspect [III and IV in Fig. 9(c)].

The terrain feature lines of the gully heads and channels exhibited a pronounced spatial migration between July 29, 2022 and March 16, 2023. At the gully heads, the migration direction of the terrain feature lines was generally opposite to that of surface horizontal deformation [see Fig. 10(d) and (e)], reflecting distinct spatial movement characteristics between geomorphic boundary migration and the horizontal deformation field. The surface horizontal deformation on the gully slopes generally showed distinct downslope movement [see Fig. 10(b)]. However, because the elevation at the bottom of the gully slope is slightly lower than that of the adjacent gully channel in this region, the surface horizontal deformation showed an opposite trend. In addition, the surface horizontal deformation on both sides of the gully slopes displayed downslope movement and converged at the gully bottom [see Fig. 10(c)].

c) Soil erosion derived from optical pixel offset-tracking and MT-InSAR: The distribution of vertical and normal deformations derived from the combined optical pixel offset-tracking and MT-InSAR was relatively dispersed [see Fig. 11(a) and (b)], with nearly the entire study catchment experiencing varying degrees of deformation. Vertical deformation varied between −17.1 and 13.9 cm, while the normal deformation ranged from −20.2 to 20.6 cm. Severe deformation was mostly distributed on gully heads and gully slopes, with a maximum value of −20.2 cm. The vertical and normal displacements showed similar patterns of spatial distribution, while the normal displacement showed more deformation points along the gully slopes. The normal deformation and the M3C2 distance revealed pronounced geomorphic changes in the study area. Both exhibited widespread negative displacements, particularly at gully slopes and gully heads [P2, P3, and P4 in Fig. 11(b) and (c)], with deformation values exceeding −10 cm and, in some areas, even surpassing −20 cm. Positive displacements were mainly distributed along the channels and at the bottoms of gully slopes. In addition, a distinct deposition was observed in the P1 area of Fig. 11(b), which was consistent with the M3C2 distance and field observations. The spatial patterns of geomorphic change identified using our method [see Fig. 11(b)] were generally consistent with those obtained from the M3C2 algorithm [see Fig. 11(c)], albeit with a loss of some fine-scale details. The geomorphic change derived from the M3C2 algorithm demonstrated a greater capacity to capture fine-scale spatial variations.

The spatial distribution of estimated erosion/deposition along with corresponding uncertainties was shown in Fig. 12. The uncertainty ranged from 3.1 to 5.6 cm, with the majority of the study area exhibiting values between 3.1 and 4.0 cm. Relatively higher uncertainties were mainly observed on steep gully slopes, whereas lower uncertainties were distributed over relatively gentle terrain [see Fig. 12(a)]. Negative values in the erosion/deposition map indicate erosion, whereas positive values indicate deposition. The subsidence areas represented by red dots (corresponding to erosion) were particularly distributed at gully heads, steep slopes, and sparsely vegetated or bare surfaces [see Figs. 12(b) and 13(a) and (b)], while deposition mainly occurred at the bottom of the slope or in areas with locally flat topography. This was consistent with field observations [see Fig. 13(c)]. Compared with 2022, the upper slope experienced severe erosion in 2023 [see Fig. 13(c)]. The locally enlarged areas I and II [see Figs. 12(b) and 13(a) and (b)] further highlighted the typical characteristics of this spatial

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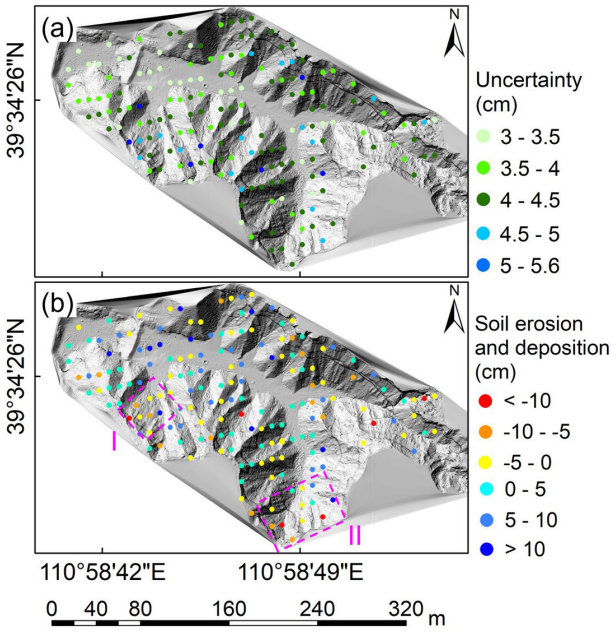


Fig. 12. Soil erosion and deposition derived from the proposed method in the Telagou catchment, including (a) spatial distribution of uncertainty and (b) cumulative erosion and deposition during July 2022 to March 2023.

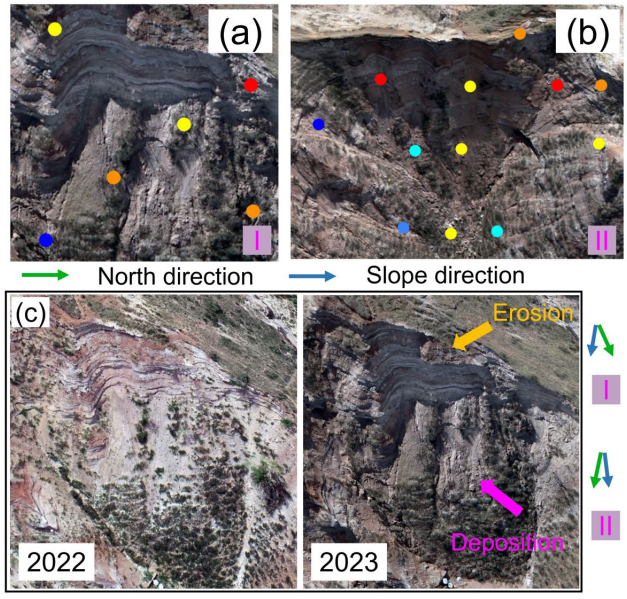


Fig. 13. (a) and (b) Enlarged views of erosion and deposition in the dotted-box areas in Fig. 12(b), (c) along with photos taken in the study area.

pattern. The locations of erosion and deposition identified by our method were generally similar to those obtained from M3C2, but lacked some of the details. Compared to erosion and deposition derived by M3C2, our results failed to capture localized erosion and deposition occurring on narrow gully slopes.

2) Assessing the Accuracy of Soil Erosion and Deposition Derived From Our Method:

a) Comparison of erosion and deposition derived from our method and M3C2: We found a linear relationship between the soil erosion derived from our method and the resampled results of the M3C2 algorithm across various parts of the slopes (see Fig. 14). The linear fitting of our method and M3C2 was close to the 1:1 line, with a significant linear correlation and relatively satisfactory NSE (R^2 = 0.61, $p < 0.01$, NSE = 0.55) [see Fig. 14(a)]. The average difference between erosion results derived from our method and the M3C2 algorithm was 0.7 cm, with maximum and minimum differences of 20.3 and −9.8 cm [see Fig. 14(b)].

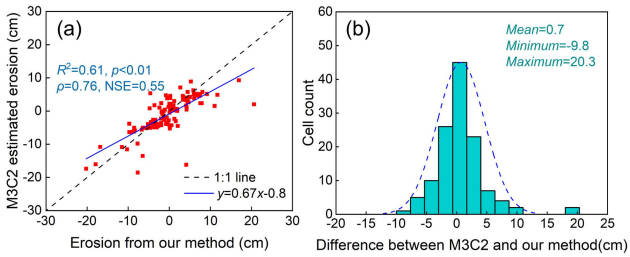


Fig. 14. (a) Correlation and (b) histogram of differences between soil erosion and deposition results derived from our method and the M3C2 algorithm, projected in the normal direction.

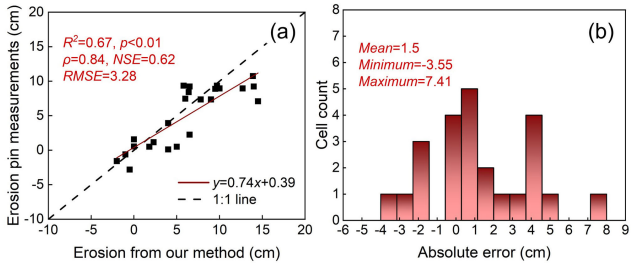


Fig. 15. (a) Correlation and (b) histogram of differences between erosion/deposition results derived from our method and the erosion pin measurements.

b) Comparison of soil erosion and deposition derived from our method and erosion pins: The absolute differences between erosion and deposition estimated by our method and elevation changes measured by erosion pins were mostly within 5 cm, as shown in Fig. 15(a) and (b). The range of differences between the two results was relatively small [see Fig. 15(b)]. The majority of differences between the two results were concentrated within the range of −2 to 2 cm, with maximum and minimum differences of 7.41 and −3.55 cm, respectively. The erosion/deposition values derived using our approach were found to correlate well and linearly with those recorded by erosion pins (R^2 = 0.67, $p < 0.01$) [see Fig. 15(a)], with the linear fit showing a good agreement with the 1:1 line. The RMSE and NSE were 3.28 and 0.62 cm, respectively.

IV. Discussions

A. Contribution of Our Study

In this study, we utilized multisource satellite-derived data and techniques (e.g., InSAR, optical POT, and UAV-P) to explore the migration of terrain feature lines and quantify the magnitude of soil erosion in the complex terrain, providing a promising framework for soil erosion monitoring over large areas. Previous studies on the optical POT have predominantly

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been applied to various applications [95], [96], [97], with high-gradient deformation. In our study, submeter-resolution imagery combined with InSAR and optical POT has been shown to be adequate for effective soil erosion monitoring.

Our study demonstrated the potential of integrating optical POT and InSAR techniques for multidimensional soil erosion monitoring in complex terrain areas using satellite remote sensing data. Compared to previous studies, our study has several advantages. First, we achieved a subpixel image co-registration accuracy within 5 cm, whereas previous studies were mostly limited to pixel-level accuracy [98]. In addition, fusion of panchromatic and multispectral bands of optical imagery using the CDFI-Net and WPCA algorithms reduced the uncertainty in optical POT monitoring and improved the reliability of monitoring. Our study combined optical POT and InSAR for soil erosion monitoring, which additionally revealed the multidimensional changes in the complex terrain [see Fig. 11(a)]. The magnitude and direction of surface horizontal deformations were obtained through multidimensional observations, which facilitated more accurate monitoring, as was shown by our results. Overall, our study achieved a high-accuracy monitoring of soil erosion in complex topographic areas and provided a new methodological framework for large-scale studies of soil erosion.

B. Registration of Remote Sensing Images

Our review of the literature revealed that registration errors caused by geometric distortions within small windows were often insufficiently addressed in complex terrain [99], [100], which constrain the accuracy of change detection results based on high-resolution imagery. Our study effectively mitigated this problem through the proposed MS-HIOF method and achieved subpixel-level registration accuracy (see Table I). Compared to Gao et al. [101], who reported that spatial losses of matched image pairs were within 1–2 pixels using the MS-HLMO method, our study met the subpixel precision requirements of high-accuracy applications.

Unlike previous studies [102], [103], the proposed MSH-IOF method achieved subpixel image co-registration. The registration errors observed in stable areas between the reference and corrected images directly validated the accuracy of our approach. The proposed method demonstrated high registration performance across different types of data (e.g., GF-7 and JL1KF01B), with the registration errors consistently within 5 cm (see Table I). Our method not only achieved a subpixel registration that accounted for subtle local geometric distortion and considered the influence of surface changes but also inherited the advantages of pixelwise displacement estimation provided by the optical flow method. Our proposed approach reduced image registration errors by 70% through the implementation of an iterative optimization strategy on top of the optical flow. In addition, compared to the research by Feng et al. [104], we introduced the least squares method to obtain the optical flow parameters for the deformation region, yielding a theoretically more reasonable result.

C. Soil Erosion/Deposition for Topographically Complex Areas

Notable ground deformation was observed by InSAR in the study catchment. However, it proved insufficient to capture detailed erosion processes in narrow gully channels due to the influence of multiple factors (e.g., image resolution, decorrelation, and radar wavelength), as shown in II of Fig. 6. Previous studies have derived vertical soil erosion solely from 1-D LOS deformation [14], [105], [106]. However, such estimates generally rely on specific assumptions or constraints, for example, assuming that horizontal deformation is negligible. Soil erosion involves the degradation of soil layer thickness. It primarily manifests as cumulative vertical deformation on flat areas, while it appears as cumulative normal deformation in steep regions. This deformation pattern challenges a single observation method like InSAR to simultaneously capture vertical displacements on flat areas and slope-normal displacements in steep slopes. Compared to Zhang et al. [17], who combined satellite imaging geometry information, LOS deformation, and topographic features to obtain vertical projections, our research achieved high-accuracy soil erosion monitoring across different geomorphic types by integrating multisource remote sensing data with multiscale observation techniques, thereby providing a reliable data basis for soil erosion/deposition studies.

We estimated soil erosion/deposition using multisource data based on InSAR and optical POT methods. Thereby we proposed the MS-HIOF method to achieve precise registration of multisource data and to provide high-quality inputs for the subsequent optical POT analysis. The proposed method demonstrated a high registration performance across different types of data. Second, the horizontal deformation derived by optical POT and co-registered JL1KF01B satellite data met our monitoring requirements. Our study demonstrated that the optical POT method was effective in capturing horizontal displacements associated with slope deformation. Notably, terrain feature line migration was observed to occur in a direction opposite to that of the horizontal deformation vectors, particularly at gully heads. These observations suggested that geomorphic boundary migration and the horizontal deformation field exhibited distinct spatial movement characteristics. Third, vertical displacements in flat areas and slope-normal displacements on steep slopes were obtained by integrating horizontal and LOS deformations. The results were in good agreement with the field erosion pins in the gully channel and the erosion/deposition derived from M3C2 on the slopes (see Figs. 11, 14, and 15). Compared with previous studies that use only one type of deformation measurement [14], our method integrated both horizontal and LOS deformations, overcoming the limitations of single data sources and thereby providing a more comprehensive representation of soil erosion/deposition processes.

D. Implications and Limitations

Our study has both scientific and practical implications, particularly given that high spatial resolution imagery has become increasingly accessible. First, the integration of SAR

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and high-resolution optical imagery enables high-accuracy soil erosion monitoring in complex terrain areas. Here, deformation across different geomorphic types in the Huangfuchuan catchment was accurately captured, which demonstrated the applicability of satellite remote sensing techniques in complex terrain and provided new insights for future studies on geomorphic change monitoring. Second, the proposed MS-HIOF method effectively addressed small-window geometric distortions under complex terrain conditions and achieved subpixel-level registration. In addition, given the coverage of satellite imagery, the combination of the MS-HIOF method with optical POT and InSAR facilitated a multidimensional and large-scale monitoring of soil erosion in the complex terrain region. Our method is not only applicable to the study area but also provides methodological support for disaster monitoring in other regions worldwide, especially in complex terrain (e.g., mountainous areas). Third, large-scale soil erosion monitoring provided a reliable basis for studying surface processes and mechanisms, which is essential for a better understanding of geomorphic evolution. Furthermore, it can also serve as a reference for practical applications associated with geological disaster prevention and risk management.

However, the application of our method to erosion still exhibits some challenges. First, the relatively coarse spatial resolution of the space-borne SAR dataset restricted the ability of the proposed method to capture detailed erosion/deposition patterns (see Fig. 6). This issue is particularly pronounced in narrow gully channels (as shown in A of Fig. 6). Second, phase unwrapping errors in the interferogram lead to an underestimation of deformation in the InSAR process [107]. In addition, severe erosion/deposition caused the deformation gradient between neighboring pixels in the interferogram to exceed half of the radar wavelength [108], preventing erosion points from being detected in the monitoring area (as shown in II of Fig. 6). Notably, the results derived from M3C2 detected significant erosion/deposition in area II in Fig. 6, whereas InSAR did not detect significant LOS deformation signals in these regions. Therefore, we call for future studies to enhance the accuracy of monitoring results by using higher quality data (e.g., long-wavelength and high-resolution SAR imagery) and adopting optimal phase unwrapping algorithms. Third, only a single ascending Sentinel-1 dataset was employed in this study, and the resulting InSAR measurements represent only the LOS projection of the actual 3-D displacement (i.e., E-W, N-S, and vertical components). Due to the side-looking imaging geometry of Sentinel-1 [32], the LOS deformation is mainly sensitive to the vertical and E-W displacement components but exhibits limited sensitivity to the N-S component. Although the integration of optical POT-derived horizontal deformation, LOS deformation, and topographic information partially mitigated the limitations associated with single-track observations, uncertainties remained in the reconstruction of the full 3-D displacement field. Future studies could integrate ascending and descending SAR observations, if possible, to further improve the reliability of deformation monitoring and quantitative assessment of soil erosion in complex terrain. Fourth, horizontal deformation obtained by the optical POT method was generally reliable, while some local areas exhibited noise. The selection of window size and step parameters during processing was usually tested through different combinations to obtain the optimal result [38], which introduced a certain degree of subjectivity. The next step is to improve the overall performance of the optical POT method and adaptively select the window size and step based on terrain characteristics to mitigate various errors. In addition, vegetation cover affects monitoring outcomes [109]. In areas with sparse vegetation (e.g., the study catchment), horizontal deformation derived from the optical POT can be reliably obtained. However, the inherent limitations of optical imagery in areas with dense vegetation present a challenge for obtaining reliable results. Due to the inability of optical sensors to adequately capture surface texture information, the quality of image matching is degraded, thereby limiting the capability of the POT method to estimate deformation in well-vegetated areas. To improve the reliability of deformation monitoring in such areas, multitrack (e.g., ascending and descending) and high-quality SAR data (e.g., NISAR data) need to be employed to obtain horizontal deformation in future studies. Furthermore, the uncertainty associated with the estimated erosion/deposition was quantified through error propagation using the covariance matrix derived from the weighted least-squares adjustment. The uncertainty propagation incorporated the uncertainties of the deformation observations, including the InSAR-derived LOS deformation and the optical POT-derived horizontal changes. This uncertainty assessment primarily reflects the propagation of measurement errors during the deformation inversion process, whereas uncertainties associated with temporal mismatches between satellite observations and field measurements were not explicitly incorporated. Although the satellite images were selected to be as close as possible to the field survey dates, a complete temporal consistency among satellite observations, UAV surveys, and erosion pin measurements was not achieved because of satellite acquisition schedules and revisit intervals. Consequently, the remaining temporal mismatch may introduce additional uncertainty into the validation results. Nevertheless, the selected datasets represented the closest temporal correspondence available for this study, thereby minimizing the influence of temporal inconsistency on the results. Future studies could further reduce this uncertainty through better coordinating field surveys with satellite acquisition schedules. Finally, this study was conducted in a naturally recovering watershed with minimal human disturbance. In other words, the observed geomorphic changes in the watershed were considered to be caused by natural factors, i.e., erosion/deposition. Future efforts could also be made to further incorporate the effects of human activities in erosion/deposition monitoring, and to utilize high-resolution time series images to quantify the velocities of erosion across different sites with variable land use types.

V. Conclusion

In this study, MCCT was proposed as a strategy for monitoring multidimensional soil erosion in the complex terrain through integrating InSAR and optical remote sensing techniques. The optical POT was used to investigate the horizontal geomorphic changes, while StaMPS-InSAR was employed to

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derive the LOS changes. These multidimensional observations were then combined to compute the magnitude of soil erosion and deposition using a weighted least-squares inversion. In order to improve the accuracy of change detection, an MSH-IOF image registration algorithm was developed to achieve subpixel registration accuracy. Along with this, high-resolution panchromatic and multispectral bands were fused to further reduce the uncertainties associated with horizontal changes. The strategy was systematically verified with field measurements and UAV remote sensing results. In addition, the uncertainty of the estimated erosion/deposition was quantified through the least-squares adjustment. The result showed that relatively higher uncertainties were mainly observed on steep gully slopes, whereas lower uncertainties were distributed over relatively gentle terrain. The combined use of InSAR and optical POT showed that severe erosion was particularly distributed on gully slopes and gully heads, which is consistent with the results derived from UAV point clouds. The erosion/deposition derived by our method was generally comparable with the results of laser scanning and erosion pins. Furthermore, the migration direction of the terrain feature lines was generally opposite to that of surface horizontal deformation at the gully heads, further demonstrating the feasibility of our results. In summary, our study represented a useful attempt for erosion monitoring on areas with complex terrain based on satellite remote sensing techniques, providing a framework for combining results of InSAR and optical imagery to derive soil erosion. However, our method is still subject to impacts induced by, for example, dense vegetation and decorrelation of InSAR results for large and rapid changes. Future efforts should be made to overcome these drawbacks and to further improve the accuracy of the monitoring results.

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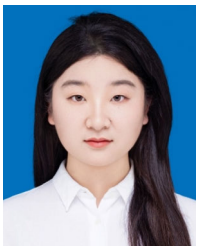
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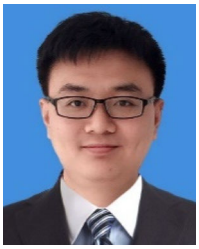
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